

Knowledge Check on Database

Database Fundamentals and Design

- ✓ What is Data?
- ✓ What is Database?
- ✓ What is Relational Database and Relational Database Management System (RDBMS)?
- ✓ How is Data stored?
- ✓ How is Data saved in MySQL?
- ✓ MySQL Storage engine and Types
- ✓ What is a Table, Column and Row?
- ✓ What is a Key?
- ✓ What is Primary Key, Foreign Key and Unique Key, Etc ...?
- ✓ What is a Transaction and ACID properties?
- ✓ Database Normalization and different forms of Database Normalization
- ✓ Create your first Database, Table and populate Table with Data

Introduction to SQL Commands

- ✓ SQL Statement and Types
- ✓ DDL Statements with examples
- ✓ DML Statements with examples
- ✓ DCL Statements with examples
- ✓ TCL Statements with examples

Query and Data Manipulation

- ✓ Create Table(s) and Temp Table(s)
- ✓ SELECT Statement in detail
- ✓ Operators, Expressions and Conditions
- ✓ WHERE Clause, ORDER BY Clause, HAVING BY Clause, GROUP BY Clause
- ✓ Select from two tables – JOINS
- ✓ Different Types of JOINS
- ✓ What is a Sub Query?
- ✓ INSERT Statement in Detail
- ✓ Load Data from CSV to MySQL
- ✓ Export Data from MySQL to CSV
- ✓ UPDATE Statement in Detail
- ✓ DELETE Statement in Detail
- ✓ What is a Stored Procedure
- ✓ What is a View?
- ✓ What is a Function?

- ✓ What is a Trigger?
- ✓ What is an INDEX?
- ✓ Clustered, NonClustered Indexes and Index Design Considerations
- ✓ INDEX Cardinality, Table Fragmentation, USE INDEX, FORCE INDEX

MySQL Backups and Recovery

- ✓ MySQL Backups, why we need backup, RTO and RPO
- ✓ Types of Backups in MySQL, Offside Storage, Encryption and Restore Testing
- ✓ Binary Logs
- ✓ The *mysqldump* Utility
- ✓ Restore of Database using *mysql* and WorkBench
- ✓ Table level Backup and Restore
- ✓ Point In Time Recovery in MySQL
- ✓ XtraBackup Utility by Percona, Functionality and Installation
- ✓ Full Backup and Restore using Percona Xtrabackup
- ✓ Incremental Backup and Restore using Percona Xtrabackup
- ✓ Schedule backups (cronjobs)

Users and Privileges Management

- ✓ Understanding MySQL Security
- ✓ Root User in MySQL
- ✓ CREATE USER and ALTER USER Statement
- ✓ MySQL Privileges, Privileges Control and Some important Privileges in MySQL
- ✓ GRANT Statement, WITH GRANT Option and Granting Privileges to MySQL User
- ✓ GRANT Tables and Checking Privileges
- ✓ REVOKE Statement and Revoking Privileges
- ✓ DROP USER Statement
- ✓ Roles in MySQL

Monitoring MySQL Server

- ✓ MySQL Server Performance Monitoring Basics
- ✓ The General Query Log
- ✓ The Slow Query Log
- ✓ The Error Log
- ✓ Optimizing *my.cnf* for MySQL performance
- ✓ InnoDB Engine Status
- ✓ Query Optimization TIPS, EXPLAIN Statement and Execution Plans

MySQL Replication

- ✓ MySQL Replication and Its Advantages
- ✓ MySQL Replication Methods, Synchronization Types and Replication Formats
- ✓ MySQL Replication Setup